

Problem. Let $G = \langle a_1, \dots, a_r \rangle$ be a free group. A word on the letters $L = \{a_1, a_1^{-1}, \dots, a_r, a_r^{-1}\}$ is called *reducible* if it equals $1 \in G$. A word is called *quasi-reduced* if it does not contain three consecutive letters $\ell\ell^{-1}\ell$ for any $\ell \in L$.

Fix a reducible quasi-reduced word $w = w_1 \cdots w_{2n}$ (with $w_i \in L$). A matching M consists of n intervals, each properly embedded in $\mathbb{R} \times \mathbb{R}_+$, satisfying the following:

- Any two intervals intersect transversally, and there are no triple points.
- $\partial M = \{1, \dots, 2n\} \times 0 \subset \mathbb{R} \times 0 \subset \mathbb{R} \times \mathbb{R}_+$.
- For each component R of $\mathbb{R} \times \mathbb{R}_+ \setminus M$, and for each component A of $M \cap \partial \overline{R}$, ∂A consists of some two points $(i, 0)$ and $(j, 0)$ with $w_i = w_j^{-1}$.

A k -crossing matching is a matching where the total number of self-intersections is k .

Let F_s be the following CW complex whose k -cells correspond to isotopy classes of k -crossing matchings. Given a k -crossing matching M with crossing set $X(M)$, the associated k -cell is given by $C(M) := \prod_{c \in X(M)} I_c$, where I_c is an interval whose endpoints correspond to the two ways of locally resolving M near c . The attaching maps are given as follows: On a facet of the form

$$a \times \prod_{d \in X(M) \setminus \{c\}} I_d, \quad c \in X(M), a \in \partial I_c,$$

let M_a be the $(k-1)$ -crossing matching obtained by resolving M at c according to a . Then the $(k-1)$ -cell associated to M_a is given by $C(M_a) = \prod_{d \in X(M_a)} I_d = \prod_{d \in X(M) \setminus \{c\}} I_d$, and the above facet maps to this $(k-1)$ -cell canonically.

Is F_s contractible?

History. A more general version of this complex is being studied by Robert Lipshitz, Lenny Ng, and Sucharit Sarkar for their upcoming paper “A stable homotopy refinement of Legendrian contact homology”, scheduled to appear in mid-2026. They have proved that F_s is always 1-connected, which is needed for some applications in their paper. If it were contractible, that would have strengthened some of the results in their paper, but much to their disappointment, they discovered that F_s is not even 2-connected in general.

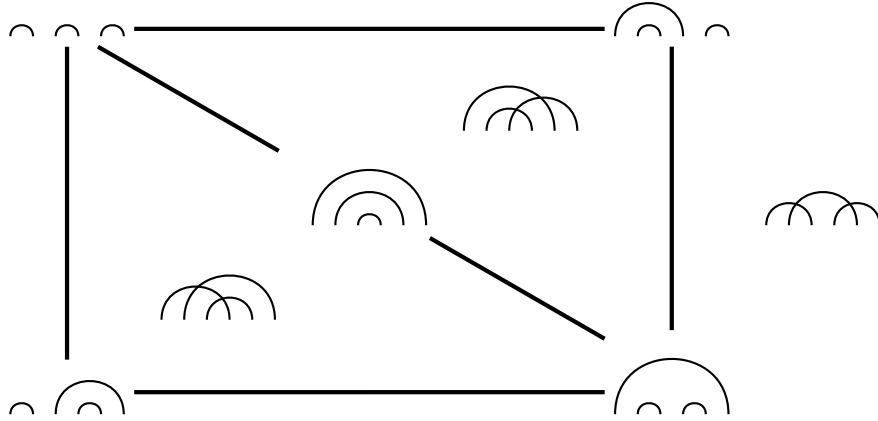


FIGURE 1. The space $F_{aa^{-1}bb^{-1}aa^{-1}cc^{-1}aa^{-1}}$. We have drawn the five 0-crossing matchings for the 0-cells and the three 2-crossing matchings for the 2-cells, but not the six 1-crossing matchings for the 1-cells.

Solution. The answer is no. The space F_s is not 2-connected in general. For instance, if $r \geq 3$ and $a = a_1, b = a_2, c = a_3$, and $w = aa^{-1}bb^{-1}aa^{-1}cc^{-1}aa^{-1}$, F_w consists of five 0-cells, six 1-cells, three 2-cells, and is homeomorphic to S^2 , as illustrated in Figure 1. $\square$